(FMEA-Z01) Portview Risks FMEA

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1. Purpose

This document records the Failure Mode and Effects Analysis (FMEA) for the Portview software, covering foreseeable hazardous situations, severity ratings, risk controls, acceptability determinations, and traceability to verification evidence.

2. Risk Scoring Method

Factor	Scale	Description
Severity (S)	1–2	1 = temporary discomfort or delayed diagnosis; 2 = re-exposure to radiation
Occurrence (O)	1–3	1 = unlikely; 2 = occasional; 3 = frequent
Risk Priority (RP)	S x O	1–2 = acceptable; 3–4 = acceptable with controls; 5–6 = unacceptable

3. Risk Identification And Control

3.1 Device Linkage And Display

ID	Title	Harm	Risk Control
`FMEA-001`	Device linkage status not visible	Delayed diagnosis	Device status indicator with real-time connection monitoring
`FMEA-002`	Viewer cannot display images	Delayed diagnosis	Image load verification with error fallback display

ID	Title	Harm	Risk Control
`FMEA-003`	Install on outside-spec PC	Delayed diagnosis	Installation requirement check and version display

3.2 Version And Audit

ID	Title	Harm	Risk Control
`FMEA-004`	Audit not possible due to incomplete logs	Temporary discomfort	Automated activity logging for all auditable operations
`FMEA-005`	Unauthorized version update	Temporary discomfort	Controlled installation with version identification

3.3 Data Security

ID	Title	Harm	Risk Control
`FMEA-006`	Leakage or forgery of transmitted data	Re-exposure to radiation	P2P-only transmission; AES-CBC encryption; network access limited to approved interfaces

3.4 Image Processing And Calibration

ID	Title	Harm	Risk Control
`FMEA-007`	Image processing module failure	Delayed diagnosis	Error detection with user-visible message; viewer fallback to unprocessed image
`FMEA-008`	Inaccurate calibration data	Delayed diagnosis	Calibration metadata preserved in TII; validated on image load

3.5 Device And Driver

ID	Title	Harm	Risk Control
`FMEA-009`	Sensor driver removed during reinstallation	Delayed diagnosis	Installation preserves driver; error message on absence
`FMEA-010`	Third-party program connection error	Temporary discomfort	Device selection UI with connection-state feedback
`FMEA-015`	Driver not installed or device connection invalid	Delayed diagnosis	Device-status monitoring with connection-state indicator

3.6 Output And Communication

ID	Title	Harm	Risk Control
`FMEA-011`	CD burning error	Temporary discomfort	Media availability check; user-visible error on failure
`FMEA-012`	Image export storage error	Service delay	Target path validation; error message on storage failure
`FMEA-013`	DICOM printer network error	Temporary discomfort	DICOM connection check; user-visible error on network failure
`FMEA-014`	Image processing error in acquired image	Delayed diagnosis	Image load validation; error display on processing failure

3.7 Annotation

ID	Title	Harm	Risk Control
`FMEA-016`	Annotations display improperly	Temporary discomfort	Annotation rendering validation; persistent storage in managed file structure

3.8 Patient Management

ID	Title	Harm	Risk Control
`FMEA-017`	Fail to open patient information	Temporary discomfort	Input validation; error message on invalid data or access failure
`FMEA-018`	Patient search failure (special characters)	Temporary discomfort	Input sanitization; controlled search query handling
`FMEA-019`	Patient information not opened despite selection	Temporary discomfort	Database record validation; error handling on open failure
`FMEA-020`	Fail to register patient	Temporary discomfort	Field validation; duplicate-check; error on registration failure

3.9 Sensor Malfunction

ID	Title	Harm	Risk Control
`FMEA-021`	Sensor malfunction prevents image acquisition	Re-exposure to radiation	Sensor error detection with user-visible message; workflow blocked until operator acknowledges; prevents repeated exposure

4. Risk Scoring And Verification

4.1 Device Linkage And Display

ID	S	O (pre)	RP (pre)	O (post)	RP (post)	Acceptable	Verification
`FMEA-001`	`1`	`2`	`2`	`1`	`1`	`yes`	`PV-TR-01` / `ITP-012`
`FMEA-002`	`1`	`2`	`2`	`1`	`1`	`yes`	`PV-TR-01` / `ITP-013`
`FMEA-003`	`1`	`1`	`1`	`1`	`1`	`yes`	`PV-SYSTR-01` / `SYSP-007`

4.2 Version And Audit

ID	S	O (pre)	RP (pre)	O (post)	RP (post)	Acceptable	Verification
`FMEA-004`	`1`	`2`	`2`	`1`	`1`	`yes`	`PV-TR-01` / `ITP-035`; Cybersecurity no 8
`FMEA-005`	`1`	`1`	`1`	`1`	`1`	`yes`	`PV-SYSTR-01` / `SYSP-007`; Cybersecurity no 3

4.3 Data Security

ID	S	O (pre)	RP (pre)	O (post)	RP (post)	Acceptable	Verification
`FMEA-006`	`2`	`2`	`4`	`1`	`2`	`yes`	Cybersecurity no 2; `PV-TR-01` / `ITP-034`; `NSE-Z01` #9

4.4 Image Processing And Calibration

ID	S	O (pre)	RP (pre)	O (post)	RP (post)	Acceptable	Verification
`FMEA-007`	`1`	`2`	`2`	`1`	`1`	`yes`	`PV-TR-01` / `ITP-013` to `ITP-021`
`FMEA-008`	`1`	`1`	`1`	`1`	`1`	`yes`	`PV-TR-01` / `ITP-008`

4.5 Device And Driver

ID	S	O (pre)	RP (pre)	O (post)	RP (post)	Acceptable	Verification
`FMEA-009`	`1`	`1`	`1`	`1`	`1`	`yes`	`PV-SYSTR-01` / `SYSP-007`
`FMEA-010`	`1`	`1`	`1`	`1`	`1`	`yes`	`PV-TR-01` / `ITP-007`

ID	S	O (pre)	RP (pre)	O (post)	RP (post)	Acceptable	Verification
`FMEA-015`	`1`	`2`	`2`	`1`	`1`	`yes`	`PV-TR-01` / `ITP-007` to `ITP-012`

4.6 Output And Communication

ID	S	O (pre)	RP (pre)	O (post)	RP (post)	Acceptable	Verification
`FMEA-011`	`1`	`1`	`1`	`1`	`1`	`yes`	`PV-TR-01` / `ITP-029`
`FMEA-012`	`1`	`1`	`1`	`1`	`1`	`yes`	`PV-TR-01` / `ITP-006`, `ITP-029`
`FMEA-013`	`1`	`2`	`2`	`1`	`1`	`yes`	`PV-TR-01` / `ITP-028`, `ITP-030`
`FMEA-014`	`1`	`2`	`2`	`1`	`1`	`yes`	`PV-TR-01` / `ITP-013` to `ITP-021`

4.7 Annotation

ID	S	O (pre)	RP (pre)	O (post)	RP (post)	Acceptable	Verification
`FMEA-016`	`1`	`1`	`1`	`1`	`1`	`yes`	`PV-TR-01` / `ITP-022` to `ITP-026`

4.8 Patient Management

ID	S	O (pre)	RP (pre)	O (post)	RP (post)	Acceptable	Verification
`FMEA-017`	`1`	`2`	`2`	`1`	`1`	`yes`	`PV-TR-01` / `ITP-002`, `ITP-004`; Cybersecurity no 1, 7
`FMEA-018`	`1`	`1`	`1`	`1`	`1`	`yes`	`PV-TR-01` / `ITP-004`; Cybersecurity no 1, 7
`FMEA-019`	`1`	`1`	`1`	`1`	`1`	`yes`	`PV-TR-01` / `ITP-002`, `ITP-004`; Cybersecurity no 1, 7

ID	S	O (pre)	RP (pre)	O (post)	RP (post)	Acceptable	Verification
`FMEA-020`	`1`	`1`	`1`	`1`	`1`	`yes`	`PV-TR-01` / `ITP-002`; Cybersecurity no 1, 7

4.9 Sensor Malfunction

ID	S	O (pre)	RP (pre)	O (post)	RP (post)	Acceptable	Verification
`FMEA-021`	`2`	`2`	`4`	`1`	`2`	`yes`	`PV-TR-01` / `ITP-008`; `PV-SYSTR-01` / `SYSP-002`

5. Cybersecurity Risk Integration

The following FMEA items have cybersecurity-related controls that are also addressed in `PV-CSRS-01` (SwRS for Cybersecurity). This cross-reference ensures that functional-safety FMEA and cybersecurity risk management are integrated.

FMEA ID	Cybersecurity Checklist Item	Risk Category	Control Integration
`FMEA-004`	Checklist no 8 (audit trail)	Detect / Respond / Recover	Activity logging provides audit trail for cybersecurity event detection
`FMEA-005`	Checklist no 3 (trusted content)	Identify / Protect	Controlled installation prevents unauthorized software updates

FMEA ID	Cybersecurity Checklist Item	Risk Category	Control Integration
`FMEA-006`	Checklist no 2 (block unauthorized network)	Identify / Protect	P2P-only transmission + AES-CBC encryption mitigate data leakage
`FMEA-017`	Checklist no 1, 7 (access control, logging)	Identify / Protect; Detect	Account management restricts access; logging enables incident detection
`FMEA-018`	Checklist no 1, 7	Identify / Protect; Detect	Input sanitization prevents injection; logging tracks search failures
`FMEA-019`	Checklist no 1, 7	Identify / Protect; Detect	Database validation prevents unauthorized access to patient records
`FMEA-020`	Checklist no 1, 7	Identify / Protect; Detect	Field validation + duplicate check prevent data integrity compromise

The cybersecurity essentials checklist in `PV-CSRS-01` Section 5.1 provides the detailed control requirements and verification mapping for these items. Network security enclosure testing (`NSE-Z01`) provides additional evidence for FMEA-006.

6. Summary

- 21 risk items identified (`FMEA-001` through `FMEA-021`)
- Severity ratings: 19 items at severity `1` , 2 items at severity `2` (`FMEA-006` , `FMEA-021`)
- Pre-control RP range: `1` to `4` ; post-control RP range: `1` to `2`
- All items assessed as acceptable after risk controls applied
- Severity `2` items (radiation re-exposure) reduced from RP `4` to RP `2` through specific controls
- All records reference Portview-controlled verification documents only